

## Comparison to Existing Tools I

## Comparison to Existing Tools II

The gap analysis established that no single tool integrates all six cross-cutting concerns. Here we synthesize the fourteen-dimension comparison into three structural insights. First, the landscape bifurcates: workflow managers (Snakemake [ @koster2012snakemake ], Nextflow [ @ditommaso2017nextflow ], CWL [ @amstutz2016cwl ]) provide distributed execution but no manuscript support; publication tools (Quarto [ @allaire2024quarto ], Jupyter Book, R Markdown [ @xie2018dynamic ], Overleaf [ @overleaf2025 ], Prism [ @openai2026prism ]) author documents but embed no integrity guarantees; and DVC [ @iterative2024dvc ] versions artifacts without orchestrating pipelines. `template/` occupies the intersection, sacrificing distributed execution for unified enforcement of testing, provenance, and documentation. This positioning is complementary—a mature deployment might use Nextflow upstream and `template/` for rendering, testing enforcement, and provenance downstream. Typst [ @madje2023typst ], with its faster compilation cycle, is not one of the nine compared peers but could serve as an alternative rendering backend if a Pandoc writer were contributed